

Nidhi Gohel

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EDUCATION

York University

Bachelor of Science in Applied Mathematics

Toronto, ON

Expected Dec 2026

- Coursework: Probability & Statistics, Stochastic Processes, Linear Algebra, Calculus, Proofs, Logic, Data Structures & Algorithms

EXPERIENCE

IKEA

Jun 2023 – Mar 2026

Returns & Exchanges Co-Worker (Data & Operations)

Toronto, ON

- Automated 5+ returns-tracking workflows with Excel macros, SharePoint and Office 365, cutting reconciliation time by 12%.
- Improved POS uptime 18% in peak windows by diagnosing and escalating incidents, cutting resolution from 55 to 45 min.
- Built Power BI dashboards on return patterns and defect categories for 3 managers, replacing a weekly manual report.

PROJECTS

Kalshi Prediction-Market Mispricing Engine | Python, Linear Programming, Team of 2

Sep 2026 – Present

- Co-building a Python engine that scans live Kalshi markets for mispricings and paper-trades them on live order books.
- Encoded no-arbitrage constraints (brackets sum to \$1, thresholds monotone) and detected violations via LP feasibility tests.
- Scanned 511 market groups, flagged 7 apparent violations, and ranked edges by fee-adjusted EV with Kelly sizing.
- Integrating an execution-cost model and backtester to compare expected edge against realized paper-trading P&L.

Stock Risk & Portfolio Analyser | Python, NumPy, SciPy, Chart.js

Jun 2026

- Simulated 500 Geometric Brownian Motion paths per asset, 90 days forward, for AAPL, GOOGL, TSLA, JPM and BTC.
- Computed 1-day VaR (95%/99%), CVaR, max drawdown, skewness, kurtosis and Sharpe ratio per asset in NumPy/SciPy.
- Built a Markowitz efficient frontier from 1,800 random portfolios to find the max-Sharpe weights, with a correlation heatmap.
- Shipped results through a Python-to-JSON pipeline into a 6-tab interactive Chart.js dashboard covering risk and allocation.

Trading Drills | JavaScript, Probability, GitHub Pages

Sep 2026

- Built a JavaScript trainer for trading interviews, hosted on GitHub Pages, with 2/5/8-minute timed mental-math sprints.
- Tracks accuracy and seconds per question across arithmetic, decimals and percentages to measure speed under time pressure.
- Designed a market-making game: quote a bid/ask (max 2 wide) on dice and coin-flip outcomes against an informed trader.
- Random customer trades act as signals, training players to update fair value under adverse selection before getting picked off.

Market Sentiment Analyzer | Python, HuggingFace Transformers, FinBERT, Scikit-learn

May 2025 – Jun 2025

- Fine-tuned FinBERT on a 5,000-headline financial news corpus, improving F1 by 12% (0.71 to 0.80) over the baseline.
- Benchmarked against a TF-IDF + logistic regression baseline and outperformed it on held-out test data.
- Cut preprocessing latency by about 40% with batched tokenization and text-cleaning pipelines for scalable NLP inference.
- Compared 3 transformer configurations on precision, recall and F1 with stratified validation to select the best checkpoint.

Media Consumption & Engagement Analytics Tool | Python, Pandas, NumPy

Dec 2025 – Apr 2026

- Found 4 statistically significant engagement patterns in 8,000+ records via frequency, duration and time-of-day analysis.
- Standardized data cleaning and transformation into modular Python scripts, replacing ad-hoc fixes that caused inconsistencies.
- Achieved 100% pipeline reproducibility across 3 independent analyst runs.
- Documented all data logic in a structured README, cutting estimated onboarding time for new contributors by 50%.

Executive Performance & KPI Analytics Dashboard | SQL, Power BI, DAX

Apr 2025 – May 2025

- Turned 10+ business questions into a self-serve Power BI dashboard with defined KPIs, cutting ad-hoc requests by about 60%.
- Designed a star-schema SQL model (2 fact tables, 5 dimension tables) to support trend and anomaly analysis.
- Implemented MoM/YoY DAX variance measures that automatically flag performance shifts of 5% or more.
- Optimized DAX measures and pre-aggregated slow dimensions in SQL views to keep dashboard load time under 3 seconds.

Trading Buddy – Stock Market Web Platform | React, Node.js, REST APIs, JWT

Jan 2024 – Apr 2024

- Served real-time data for 50+ tickers from 2 financial REST APIs through a Node.js backend at under 500 ms average latency.
- Implemented JWT authentication with token expiry and refresh to secure user sessions across multi-page navigation.
- Built discussion threads and interactive financial guides, increasing the user engagement surface by 3x.
- Architected a modular React UI so new features can be added without code rewrites.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript

Quantitative: Probability, Stochastic Processes, Statistical Testing, Monte Carlo Simulation, VaR/CVaR, Portfolio Optimization, Linear Programming, Kelly Criterion

Data Science & ML: Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, HuggingFace Transformers, FinBERT, LangChain, NLP, A/B Testing

Software Engineering: React, Node.js, Next.js, Three.js, REST APIs, JWT, HTML/CSS, jQuery, Git, GitHub, Jupyter, VS Code

Databases & BI: PostgreSQL, MySQL, Star-Schema Modeling, Power BI, DAX, Excel, SharePoint, Office 365